11

SOUND VARIATION IN EUROPEAN PORTUGUESE

Celeste Rodrigues

This chapter describes some of the most relevant variable phonological processes of European Portuguese (EP) applying at the word level. We begin by introducing variable phenomena in EP, as well as the properties of its standard variety, including specific variable rules, such as the behavior of sonorant /ɾ/ in onset and coda position, the fricative /S/ in syllable coda, word-initial unstressed coronal vowels, and stressed coronal vowels before palatal segments. Next, we analyze the word-level phonological phenomena most often referred to as northern EP linguistic markers. We describe phonological alternatives to certain segments of standard EP (namely, /β/, /ʧ/, and apical fricatives) and other geographically restricted phenomena that have not received much attention in the literature, such as the development of secondary consonant articulations in stressed syllables. Next, we describe other phonological phenomena, such as the simplification of diphthongs, the palatalization of /u/, and properties of the vowel system from the Alentejo and Algarve varieties. Furthermore, we address phonological phenomena from the varieties spoken in Madeira and the Azores, including challenging ones, such as specific processes undergone by stressed vowels, the fricative /S/, and the vowel chain shift in the speech of São Miguel Island, among others. Although more geographically restricted, these phenomena show particularities of Portuguese phonology that deserve attention.

1 Introduction

The present chapter describes some of the most important variable phonological phenomena of European Portuguese (EP) at the word level.[1] We begin by defining phonological variation, a phenomenon whereby a given word structure occurs in speech with more than one output, recognized as such by the speakers of the language or proven by acoustic and/or perceptual analyses. Different outputs of underlying structures may occur in a language depending on, among others, linguistic, social, stylistic, and individual factors. The distinct outputs may emerge only in some varieties (or dialects) of the language and may be restricted to some social groups or speakers. Some may be interpreted as phonological, and others may reflect only minor phonetic differences. In this sense, variable phenomena differ from alternation phenomena, which only depend on internal linguistic factors, such as the environment where the segment occurs.

DOI: 10.4324/9781003266648-14

Phonological variable phenomena may be of two types, depending on whether they involve the same or different phonological forms in the dialects of a language. To exemplify the former group of phenomena, we may state that: i) all dialects of EP have two allophonic variants of word-initial unstressed /e/ (i.e., speakers either pronounce [i] or [e] in words such as *elefante* 'elephant' and *emissão* 'emission', cf. Rodrigues (2003), apparently without a specific reason for their choice); ii) there are word-final allophonic variants of the coronal fricative /S/ (i.e., some speakers may produce [ʃ], [ʒ], or [z] regularly – Mateus & Andrade, E. d' 2000, depending on the segmental context, while other speakers follow a new pattern of variant distribution across contexts); and iii) there are several phonetic forms for stressed /e/ before [–anterior] (also called palatal) segments (i.e., speakers produce [e], [ej], [ɐj], [ɐ], or [a] – Barros 1994), whether both segments belong to the same syllable (*ameixa* 'plum') or they do not (*cereja* 'cherry').[2]

The second type of variable phonological phenomena concerns, for instance, the production of the first segment in the word *voz* 'voice'. In this case, speakers of different EP dialects do not share the same phonological representation—we argue that they have either labiodental /v/ or bilabial /β/ in the underlying form, simplifying, then, the analysis provided by Rodrigues 2003, which derived [β] from /v/ in a northern dialect. Another example involves apical vs. coronal fricatives, which are also distributed differently throughout Portugal's territory (Segura 2013). Both types of variable processes—either with the same or with different phonological representations in the dialects of a language—will be discussed in the remaining part of this chapter. The depth of the following discussion reflects the asymmetry between the higher number of dialectal descriptions of EP versus its sociolinguistic descriptions. Thus, we aim to draw scholars' attention to several phenomena of EP that have received unequal focus but which nonetheless merit further investigation, such as the special features from vowel and consonant segments, types of competing phonological phenomena, their greater or lesser chances of survival in EP dialects, and the interaction among phonological phenomena.

While not all theoretical perspectives support the need for the exact formulation of a phonological representation from which all possible phonetic representations of its constituents are derived—whether we think of phonological segments in a generative perspective or elements in models like Element Phonology, Government Phonology, Particle Phonology, and so on (cf. Raimy & Cairns 2015) or even in an Optimality-theoretic perspective—it is nonetheless advantageous to consider here an abstract representation for convenience. The nature of underlying forms can be inferred by observation of common properties of phonetic units that realize it in most cases or by reasons of the descriptive tradition of the language, sometimes related to the alphabetic writing system. This underlying representation has the core features that constitute the speakers' general production target, although, in spontaneous speech, it may be subject to different phonetic implementations arising from the speaker's profile, the discourse situation where he is involved, or the social group to which he belongs. This chapter does not regard the question of whether this representation is already present in speakers' mental grammar or whether it is built from their experience with language use. The goal here is to determine the unavoidable properties of the abstract sound units of EP that are manipulated by speakers in current phonological processes. Although we cannot avoid using terms like 'rule' and 'process' to refer to the phonological phenomena addressed, we do not assume a particular view on the nature of the grammar besides the existence of, at least, two levels of representation of speech, namely, the underlying representation and the phonetic representation. Thus, we discuss phonological variation in the EP sound system considering its several varieties, not only in mainland Portugal, but also in the islands of Madeira and the Azores. Before addressing individual phenomena, however, we summarize the properties of the Standard

European Portuguese (SEP) spoken in the Lisbon metropolitan area (cf. also Chapters 1 and 2 in this volume).

Despite minor differences in interpretation, most linguists working on Portuguese agree that the sound system of SEP includes seven distinctive oral vowels (i.e., /i, e, ɛ, a, u, o, ɔ/) and 21 consonants (i.e., /p, b, t, d, k, g, f, v, s, z, ʃ, ʒ, m, n, ɲ, l, ʎ, ɾ, kʷ, gʷ, k͡s/ (Andrade, A. 2020; Andrade, E. d' 1977; Freitas 2001; Mateus 1975; Mateus & Andrade, E. d' 2000; Rodrigues, C. 2020; Vigário 2003). All vowels may occur individually as the nucleus of stressed syllables, and all consonants may appear in the onset position. Only /ɾ/, /l/ (produced with velarization) and /S/ may figure as syllable codas in SEP, with /S/ being the underlying form of the fricative in syllable codas. Liquids /ɾ/ and /l/ may combine with obstruent consonants in branched syllabic onsets, such as /bɾ/ (e.g., *cobra* 'snake'), /kɾ/ (e.g., *crer* 'believe'), /fɾ/ (e.g., *fraco* 'weak'), /fl/ (e.g., *flor* 'flower'), /pl/ (e.g., *planta* 'plant'), /tl/ (e.g., *atlas* 'atlas'). Similarly, vowels may combine in branched nuclei (i.e., falling diphthongs, phonetically), such as /eI/ (e.g., *leite* 'milk'), /oI/ (e.g., *noite* 'night'), /aU/ (e.g., *pauta* 'agenda'), /aI/ (e.g., *pai* 'father'). Rising diphthongs emerge phonetically if the structure contains vowels in hiatus (e.g., *sério* 'serious', *piano* 'piano'). Within the syllable nucleus, vowels and falling diphthongs may undergo nasalization, an assimilation process due to the presence of a nasal auto-segment /N/, as in *planta* 'plant', *conde* 'count', *cinco* 'five', *mundo* 'world', *irmã* 'sister', *fim* 'end', *irmão* 'brother', *cães* 'dogs'.

There is disagreement among EP scholars concerning the number of rhotic sounds (i.e. some consider /ʀ/ a phonological unit, apart from /ɾ/, while others do not—see discussion below), the number of vowels (i.e., some include /ɐ/, /ɨ/[3]—see Veloso 2013, 2016, and nasalized vowels as phonological units, while others do not), and the inclusion (or not) of glides [j] and [w] in the phonological system of SEP. Our proposal follows the rather restrictive lines of Mateus and Andrade, E. d' (2000), namely, we sustain that underlying segments are, exclusively, the unpredictable units that emerge within structures with no other regular explanation. In this way, [ʀ], [ɐ], [ɨ], the glides [j] and [w], and nasalized vowels all emerge in specific contexts as instances of other phonological units rather than having phonological status themselves (i.e., /ʀ/, /ɐ/, /ɨ/, etc.). The coronal sonorant /ɾ/ may surface as [ʀ] due to some prosodic restrictions (e.g., in the word-root initial position, and in a syllable onset position after a coda consonant or the auto-segment /N/), /a/ surfaces as [ɐ] in unstressed and nasalized nuclei, while /e/ surfaces as [ɐ] in stressed contexts and as [ɨ] in unstressed syllables only.

Unlike alternation phenomena that depend on linguistic factors exclusively, most phonological variable phenomena of EP concern the phonetic shape of phonological segments that arise within or across dialects due to individual, social, or stylistic factors, apart from the linguistic factors. Dialectology, above all areas of linguistic studies, plays a major role in identifying sound variants of phonological segments of EP, even if a complete phonological analysis of the facts is not always pursued in dialectal literature. Some processes apply only to specific structures, either word-internally or between words, while others apply more generally, although some of them are restricted to specific varieties of the language. Variation phenomena in EP have not been studied deeply within a social dimension. Therefore, we may provide only a rough picture of them here.

In what follows, we refer occasionally to the auto-segmental framework (cf. Goldsmith 1995), which has been accepted by most EP phonologists after the phonological description of SEP presented in Mateus and Andrade, E. d' (2000).

This chapter is organized as follows. Section 2 describes the properties of phonological units of SEP. Section 3 presents the phonological traits that are often used to distinguish regional EP dialects within mainland Portugal. Section 4 complements section 3 with an analysis of less studied

processes found in several EP varieties, not only describing these processes but also determining their scope. Section 5 provides a summary and concluding remarks.

2 The phonology of SEP

2.1 Oral vowels and diphthongs in SEP

SEP contrasts oral vowels /i, e, ɛ, a, u, o, ɔ/ in stressed syllables, as previously mentioned,[4] as illustrated in (1). The number of words with the contrasts /e/ vs. /ɛ/ and /o/ vs. /ɔ/ is lower than the rest of vowel contrasts in the language.

(1)

/i/ vs. /e/	[i] – *pira* 'pyre'	[e] – *pera* 'pear'
/e/ vs. /ɛ/	[e] – *seca* 'dry'	[ɛ] – *seca* 'dries'
/a/ vs. /ɔ/	[a] – *sala* 'room'	[ɔ] – *sola* 'sole'
/o/ vs. /ɔ/	[o] – *tola* 'silly'	[ɔ] – *tola* 'head'
/o/ vs. /u/	[o] – *soco* 'punch'	[u] – *suco* 'juice'

Despite this general behavior, some oral vowels may not surface as expected in stressed syllables. This is the case of /e/ and /ɛ/, which may surface, respectively, as [ɐ] or [e] and [ɐ] or [ɛ], either before heterosyllabic palatal consonants /ʃ, ʒ, ɲ, ʎ/ (e.g., *seja* 'be', *senha* 'password', *telha* 'tile', *flecha* 'arrow') or before the palatal glide [j] syllable-internally (e.g., *leite* 'milk', Barros 1994). Thus, coronal mid vowels /e/ and /ɛ/ may become dorsal ([+back]), i.e., [ɐ], if they are followed by a coronal ([–anterior]) consonant. If we accept the underspecification of vowels (cf. Archangeli 1988) used in Mateus and Andrade, E. d' (2000), /e/ and /ɛ/ do not need to be specified in the underlying form for place node (V-Place). Both of these mid vowels undergo, then, a complementary specification process before consonants, as illustrated in Figure 11.1.

This process may also apply before a fricative coda in certain words (e.g., *sexta* 'sixth', *texto* 'text'), but not in other words with a similar context, which are excluded due to etymological reasons (e.g., *cesta* 'basket', *vesga* 'cross-eyed', *festa* 'party'). Although the specification process

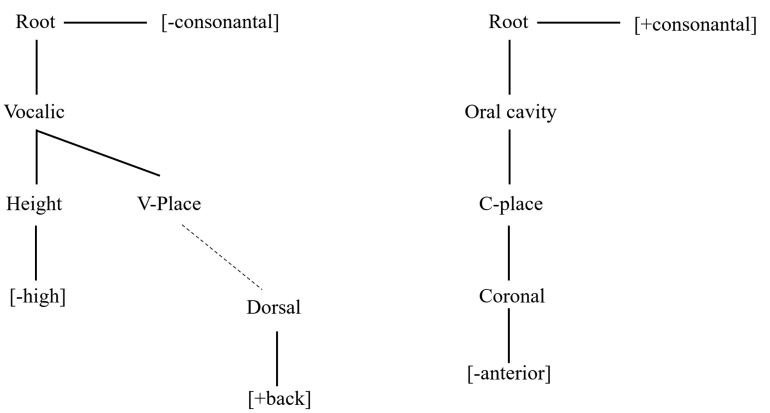

Figure 11.1 /e/ dorsal specification.

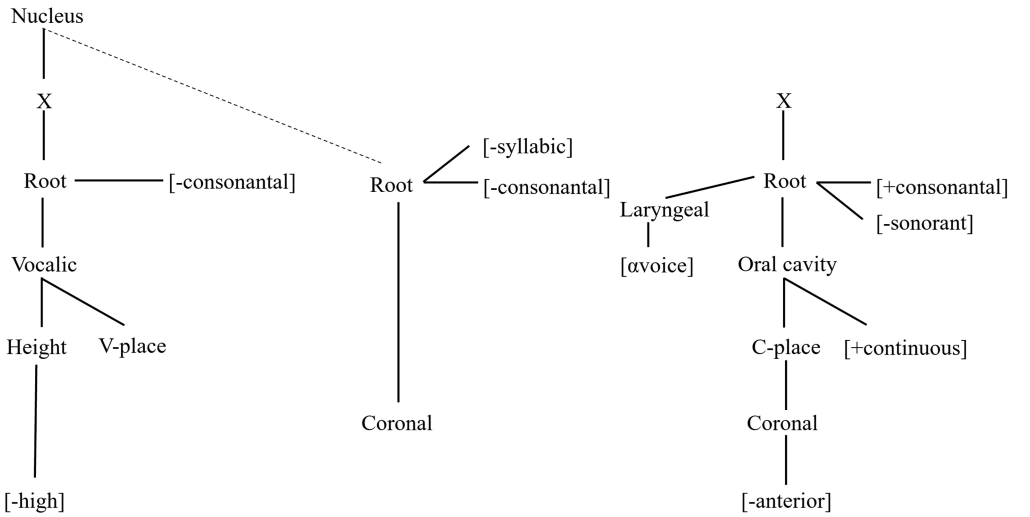

Figure 11.2 Coronal glide insertion before [–anterior] consonants.

may apply to both /e/ and /ɛ/, it is more generally applied to words with /e/ (see Barros 1994 for a study on /e/ and /ɛ/ before palatal segments in the Lisbon variety). Furthermore, some words with /e/ may also undergo a diphthongization process resulting in [ɐj], as in *fecha* 'close', *seja* 'be', *sexto* 'sixth'. Diphthongization of /e/ occurs, exclusively, before fricative sounds [ʃ] or [ʒ] in SEP—not before palatal sonorants as it may happen in Northern dialects (see Rodrigues, C. 2003 for a study comparing the Lisbon and Braga dialects). Diphthongization of /e/ involves both the syllabic and the melodic tiers, as shown in Figure 11.2.

In the branched /eI/ nuclei, which surface as [ɐj] in SEP, a dissimilation process has changed the coronal vowel into a dorsal one, as illustrated in Figure 11.3.

Unstressed syllables do not keep all vowel contrasts phonetically, unlike stressed ones. Indeed, /e/ and /ɛ/ are regularly produced as [ɨ], not only in SEP but also in all varieties of EP (as in *sedoso* 'silky', *festival* 'festival'), while /o/ and /ɔ/ are produced as [u] (as in *lobinho* 'little wolf', *colado* 'pasted'), and the vowel /a/ is produced as [ɐ] (as in *acabar* 'to finish'). There are, however, some exceptions to the reduction processes affecting the phonological vowels in unstressed syllables, such as *aquecer* 'to heat', *Alcácer* (toponym), and *mestrado* 'master's degree' with [ɛ]; *adotar* 'to adopt' with [ɔ]; and *âmbar* 'amber' and *ação* 'action', which surface with [a]. Other exceptions occur in word-initial position and are subject to variation in SEP. Word-initial /e/ may surface as [e] and as [i], if it is subject to a rising rule (e.g., *elevador* 'lift', *emigrante* 'emigrant'); /ɛ/ surfaces as [ɛ] (e.g., *ervinha* 'small herb'), while /o/ surfaces as [o] and [ɔ], due to a lowering-rule (e.g., *origem* 'origin', *ocasião* 'occasion'). Mascarenhas (1996) reports that, with regard to word-initial unstressed /o/, young speakers with high socioeconomic status of Lisbon and Porto prefer [ɔ] over [o].

The high vowels /i/ and /u/ correspond to phonetic [i] and [u] in unstressed syllables, although some, if adjacent to another vowel, may become glides or be deleted in fast speech (Rodrigues 2003). This implies that the labial vowels /ɔ, o, u/ all display [u] in unstressed syllables, as we observe when comparing vowels from morphologically related words, e.g., *sol* [ˈsɔɫ] 'sun' / *enso-*

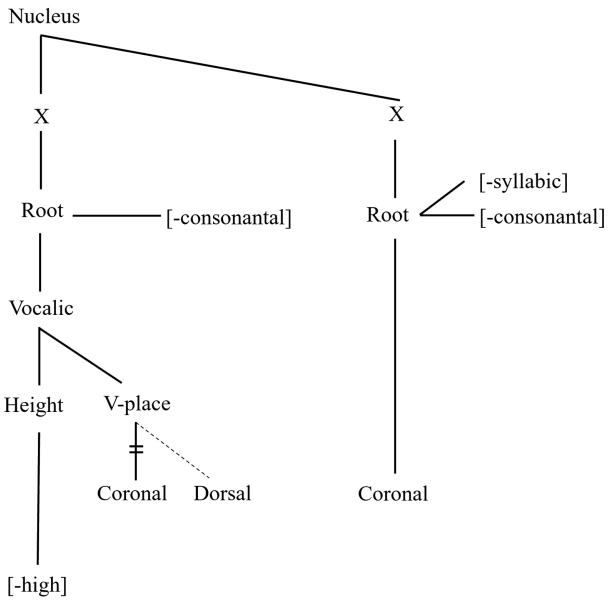

Figure 11.3 /e/ dorsalization before the coronal glide.

larado [ẽsulɐˈɾadu] ‘sunny’, *bolor* [buˈloɾ] ‘mold’ / *bolorento* [buluˈɾẽtu] ‘mouldy’, and *curva* [ˈkuɾvɐ] ‘curve’ / *curvado* [kuɾˈvadu] ‘curved’. Deletion of [ɨ] and [u] is a common process not only of Lisbon speech but also of several other EP varieties. This process applies not only to unstressed vowels in word-final position (e.g., *ponte* ‘bridge’, *ponto* ‘point’), but also word-internally if the vowel corresponds to an underlying non-high vowel (e.g., *costume* [kʃˈtumɨ] ‘custom’, *deturpado* [dtuɾˈpadu] ‘misrepresented’). There are differences of frequency, however. Rodrigues, C. (2003: 295–297) and Rodrigues, C. (2016) report that [ɨ] and [u] are more often deleted if they correspond to non-high vowels than when they correspond to underlying high vowels. If a word-final non-high vowel is followed by a consonant of a following word, deletion is the most common result, as in *gato preto* [ˈgatˈpɾetu] ‘black cat’, *teve tudo* [ˈtevˈtudu] ‘had all’. The same process takes place in word-final syllables containing /S/, as in *gatos* [ˈgatʃ] ‘cats’, *espertos* [ʃˈpɛɾtʃ] ‘smart’, *verdes* [ˈveɾdʃ] ‘green’, *cores* [ˈkoɾʃ] ‘colors’. [ɨ]-deletion is common in word-internal CV syllables (not as much in CVC syllables), while [ɐ] is kept word-finally, as well as word-internally.

Overall, this means that oral coronal and labial non-high vowels are phonetically stable in stressed syllables only, unlike high vowels, which also keep their underlying features in unstressed syllables phonetically. Unstressed non-high vowels /e/, /a/, and /o/ surface, respectively, as [ɨ], [ɐ], and [u], depending on their phonological articulation place word-internally. At the beginning of words—as already mentioned, unstressed coronal and labial vowels do not undergo the same rules, surfacing either as [i] / [e] or [o] / [ɔ], e.g., *emancipar* ‘to emancipate’, *orelha* ‘ear’.

There are several oral falling diphthongs phonetically in SEP besides [ɐj], such as, [aj] *pai* ‘father’, [aw] *mau* ‘bad’, [iw] *riu* ‘he/she laughed’, [uj] *fui* ‘I went’, [oj] *noite* ‘night’, [ɔj] *dezoito* ‘eighteen’, [ɛj] *pincéis* ‘brushes’, and [ew] *eu* ‘I’. The second vowels in these diphthongs correspond to underlying /U/ or /I/, due to its special status concerning syllable nuclei creation and

stress assignment. Some of these diphthongs derive from two adjacent underlying vowels synchronically, but others derive from a VC sequence historically (e.g., [ɛj] in *pincéis* 'brushes', which is the plural form of *pincel* 'brush', cf. Chapter 14).

There is a special diphthong in EP, namely, [ow] (e.g., *mouro* 'muslin', *andou* 'he/she walked'), which historically underwent a monophthongization process in SEP (and also in other Southern varieties of EP). This diphthong surfaces as [o], due to a change rule that started to apply in the 17th century in Central and Southern dialects of EP (Teyssier 1980, cf. Chapter 14).

2.2 Vowel and diphthong nasalization in SEP

One major property of EP is that phonological vowels undergo a nasalization process in syllables of words derived from Latin that contained a nasal coda consonant, corresponding nowadays to the auto-segment /N/ (in other words, an un-associated [+nasal] feature). See, for instance, the words *campo* 'field', *conde* 'count', *quente* 'hot', *cinco* 'five', and *fundo* 'bottom', all containing the phonetic nasal sounds [ɐ̃], [õ], [ẽ], [ĩ], and [ũ], respectively. Most nasalized vowels also emerge in stressed syllables word-finally, e.g., *irmã* 'sister', *som* 'sound', *fim* 'end', *atum* 'tuna'. The coronal vowel /e/, however, undergoes a complementary diphthongization process at the end of the word, regardless of word stress, e.g., *Belém* (toponym) and *batem* 'they hit' both end in a nasal diphthong [ɐ̃j̃]. This means that the coronal vowel in word-final position surfaces as [ɐ̃], a dorsal nasal vowel, followed by a coronal glide (a consequence of [j]-insertion). Thus, after the insertion of the coronal glide, /e/ becomes dorsal due to a dissimilation process, as shown in Figure 11.3.

Nasalization also affects vowels in unstressed syllables, as in *ventania* 'windstorm', *companhia* 'company', *andar* 'to walk', *pintar* 'to paint', *juntar* 'to join'. Word-final unstressed /aN/, besides changing from [+low] to [-low], diphthongizes and nasalizes, surfacing as [ɐ̃w̃]. This process can be seen in verb forms such as *falam* 'they speak' and *choram* 'they cry', where the insertion of the labial glide [w] occurred (notice that the insertion is in accordance with the [+back] articulation feature of the vowel), creating the phonetic diphthong [ɐ̃w̃]. A similar process happens when unstressed /e/ changes to [ɐ̃j̃], as in *sabem* 'they know' and, also, in words where [e] is derived historically from /i/, such as *partem* 'they leave', from the verb *partir* 'to leave' (Mateus & Andrade, E. d' 2000).

SEP, unlike other dialects of EP (cf. Lao 2020), does not nasalize vowels before heterosyllabic nasal consonants, as in *lama* 'mud', *cana* 'cane', *apanha* 'he catches', all with stressed [ɐ]. We also observe this in words such as *falamos* 'we speak' (with stressed [ɐ]) and *falámos* 'we spoke' (with stressed [a]), in which the stressed vowel is not nasalized. Moreover, morphologically underived words such as *lua* 'moon' and *coroa* 'crown' surface with a vowel hiatus without nasalization in EP, although the following syllable had /n/ in Latin (< LUNA, CORONA). According to Mateus and Andrade, E. d' (2000), this is a consequence of the non-association of /N/ of this morphological class of words. Only morphologically derived words, e.g., *lunático* 'lunatic', *coronal* 'coronal', are produced with [n] fulfilling the following empty onset position—a consequence of the adjunction of the derivational morpheme.

2.3 Variable consonantal rules of SEP

Most phonological consonants of SEP are contrastive in minimum word-pairs, as shown in (2).

(2)

/p/ vs. /b/	[p] – *pala* ‘visor’	[b] – *bala* ‘bullet’
/t/ vs. /d/	[t] – *toca* ‘den’	[d] – *doca* ‘dock’
/k/ vs. /g/	[k] – *calo* ‘callus’	[g] – *galo* ‘rooster’
/f/ vs. /v/	[f] – *fala* ‘speech’	[v] – *vala* ‘ditch’
/s/ vs. /z/	[s] – *cinco* ‘five’	[z] – *zinco* ‘zinc’
/ʃ/ vs. /ʒ/	[ʃ] – *chá* ‘tea’	[ʒ] – *já* ‘now’
/m/ vs. /n/	[m] – *mó* ‘millstone’	[n] – *nó* ‘node’
/n/ vs. /ɲ/	[n] – *mana* ‘sister’	[ɲ] – *manha* ‘guile’
/l/ vs. /ʎ/	[l] – *fila* ‘line’	[ʎ] – *filha* ‘daughter’
/l/ vs. /ɾ/	[l] – *calo* ‘callus’	[ɾ] – *caro* ‘expensive’

Other phonemes include consonants with secondary labial articulation, namely, /kʷ/ and /gʷ/, especially in words with Latin and Greek roots (e.g., *qualidade* ‘quality’, *aquático* ‘aquatic’, *água* ‘water’, *guarda* ‘guard’, *igual* ‘equal’, cf. Freitas 2001) and some recent borrowings from modern languages. Native speakers do not treat them as productive segments of Portuguese nowadays, using them only in a limited set of words. The same happens with /k͡s/, as in *óxido* ‘oxide’, *fluxo* ‘flow’, *fénix* ‘phoenix’ from Greek and Latin, and *pirex* ‘pyrex’ and *táxi* ‘taxi’ from modern languages.

SEP consonants do not show considerable variation, although we must concentrate on a few aspects of their phonology. The first issue we want to mention concerns the sonorant /ɾ/ or, rather, the rhotic consonant /ɾ/ and /ʀ/, depending on the number of underlying segments we consider. The second issue deals with the phonetic nature of coda-/S/, and the third is related to the voiced stops /b/, /d/, and /g/.

Starting with rhotic consonants, the disagreement on the number of rhotics in Portuguese is mostly due to their behavior in intervocalic position, since in word-initial position it is commonly assumed that the uvular trill [ʀ] results from a prosodic constraint (Vigário 2003). Words like *caro* ‘expensive’ have [ɾ] in SEP, while words like *carro* ‘car’ have [ʀ] (among other variants). This phonetic difference may be viewed as an underlying contrast between two segments or, instead, as one segment with different allophones. However, this contrast is exclusive of some specific contexts. In other words, it is a weaker contrast than the other phonological contrasts in the language (cf. Pereira, R. 2020 for more details). There are contexts like word-initial (e.g., *rosa* ‘rose’) and post-coda position (e.g., *Israel* ‘Israel’, *melro* ‘blackbird’) where [ʀ] is expected in SEP. Following a nasal vowel, the rhotic must be produced as [ʀ] (e.g., *tenro* ‘tender’). Elsewhere, the rhotic must be produced as [ɾ], e.g., *sorte* ‘luck’, *prato* ‘dish’, *dinossauro* ‘dinosaur’, *pira* ‘pyre’, or deleted if word-final as in *dor* ‘pain’ (specially, in the end of an intonational phrase).

Some phonologists (Mateus & Andrade, E. d’ 2000; Rodrigues, C. 2003; Vigário 2003) claim that, despite all the phonetic variation it allows, /ɾ/ is the only underlying rhotic of EP, since the allophone [ʀ] only surfaces after a syllable coda (e.g., *melro* ‘blackbird’, *Israel* ‘Israel’), nasalized vowels (*tenro* ‘tender’) or as the result of a strengthening process in word-initial position (e.g., *rato* ‘mouse’). This analysis faces several problems, most of which derive from the phonetic variation that the segment allows. One problem concerns the explanation for words like *carro* ‘car’, for which the aforementioned authors claim that there are two adjacent rhotics in the underlying form, i.e., one in coda and the other in the following onset. This coda would have to be furtherly syllabified as the onset of the second syllable of the word, giving rise to a geminate [r], which is still a phonetic variant found across Portugal, the Lisbon region being the apparent exception.

Since processes tend to involve segmental classes and not segments individually, this analysis seems correct because it has the advantage of joining all coda-consonants constraining the rule. The so-called strong-R emerges, then, after all segments that behave like codas in the language; otherwise, it is derived from a strengthening rule word-initially or derivational morpheme-initially (cf. Vigário 2003).

Rhotic sounds currently undergo considerable variation in EP in all contexts—an issue never clearly addressed by the aforementioned authors—as described in Rodrigues, S. (2015) and Pereira, R. (2020). In fact, [ʀ] seems to be a recent variant that emerged in the history of Portuguese only during the 19th century (cf. Chapter 14). It was at first considered an error of production, but it eventually spread out and became the predominant variant in the Lisbon metropolitan area (Rodrigues, C. 2003). Currently it has different phonetic manifestations in different EP areas (cf. Rennicke & Martins, P. T. 2013, Rodrigues, S. 2015), including anterior and posterior variants (for instance, in insular EP, Pereira, R. 2020 found a great number of posterior phonetic variants). SEP has changed the former alveolar trill [r] (which nonetheless still prevails elsewhere in the country, cf. Pereira, R. 2020) into a dorsal segment [ʀ], which is nowadays spreading toward several urban areas. Considerable phonetic variation can be found. Some sounds keep the sonorant features of the rhotic, while others do not, such as the uvular fricative [ʁ].

Several other phonologists and phoneticians, however, consider the fine-grained properties of rhotic sounds to be more important than their global phonological distribution to characterize the underlying system (e.g., Jesus & Shadle 2005; Rennicke & Martins, P. T. 2013; Rodrigues, S. 2015; Veloso 2015, 2019). To them, the existence of two phonological units, namely /ʀ/ and /ɾ/, explains the data in a more satisfactory manner. These authors claim that /ɾ/ and /ʀ/ have phonetic variation of their own and, additionally, that there is not enough evidence to postulate that the uvular trill still is the phonological segment of EP nowadays in words like *carro* 'car', since it is more often produced as an obstruent (fricative) than as a sonorant in the data they studied. Rennicke and Martins, P. T. (2013) question whether the strong-R is not already an underlying /ʁ/. Acoustic analysis suggests the same (Rodrigues, S. 2015). Further, Pereira, R. (2020) claims that, even though the trill [r] is found in all regions of Portugal except the Lisbon metropolitan area, it cannot be understood as a product derived from two taps, since taps are intrinsically different from trills. For that reason, Pereira, R. (2020) concludes that the phonological analysis of Mateus and Andrade, E. d's (2000) is not the most adequate. Given the complexity of this issue, the reader would benefit from consulting other references, such as Bonet and Mascaró (1997), Veloso (2019), and the literature review on rhotics in Pereira, R. (2020), which considers phonological, historical, dialectal, and acquisition data. Pereira, R. (2020) concludes that although [r] predominates in several varieties of EP, posterior phonetic segments predominate in the Lisbon and insular areas, in addition to some isolated places in Central and Northern Portugal.

The most adequate interpretation of the variation data of the strong-R in EP (be it a /ɾ/ or a /ʀ/) acknowledges that it has been a phenomenon of stable variation throughout the history of EP (Pereira, R. 2020: 159), involving anterior and posterior segments, sometimes sonorants and sometimes fricatives. The weak contrast between two units occurs only intervocalically. This means that, in most of the contexts where posterior variants emerge, they stand for /ɾ/, namely word-initially (or morpheme-initially) and after coda consonants.

Besides the variation associated to /ɾ/ in onset and coda contexts, Andrade, E. d' & Viana (1994) claim that a word-final coda /ɾ/ may delete before [r] if speakers use this phonetic variant as the onset rhotic in *ir rapidamente* 'to go rapidly', but not before [ʀ] if this is their choice. Rodrigues, C. (2003: 244–258) reports on data from /ɾ/-deletion in the speech of different groups of speakers in Lisbon and Braga (cf. also Mateus & Rodrigues, C. 2004) and finds that /ɾ/ may also delete in

$$\begin{bmatrix} \text{-sonorant} \\ \text{-continuous} \\ \text{+voice} \end{bmatrix} \longrightarrow \begin{bmatrix} \text{+continuous} \\ \text{-strident} \end{bmatrix} \;/\; \left\{ \begin{matrix} [\text{-sonorant}] \\ \begin{bmatrix} \text{+sonorant} \\ \langle \text{-continuous} \rangle \\ \alpha\text{coronal} \end{bmatrix} \end{matrix} \right\} (\#) \; [\, < \overline{\text{-}\alpha\text{coronal}} > \,]$$

Figure 11.4 Spirantization rule of voiced stops in EP (Andrade, E. d' 1977: 189).

other contexts in Lisbon, namely, before vowels (e.g., *ter amigos* 'to have friends'), all classes of consonants (e.g., *ter medo* 'to have fear', *mar salgado* 'salted sea'), and even in sentence-final position (e.g., *Eu vou ler.* 'I am going to read.'). However, in the latter context, insertion of [ɨ] after the rhotic prevails. In sandhi, deletion of /ɾ/ occurs more often before another /ɾ/ than in any other context.[5]

With regard to coda-/S/, SEP exhibits a pattern of complementary distribution among the segments [ʃ], [ʒ], and [z], both within the word and between words (Andrade, E. d' & Viana 1993). [ʃ] emerges before voiceless consonants (e.g., *descida* 'downhill', *três carros* 'three cars'), [ʒ] emerges before voiced consonants (e.g., *desgraça* 'misfortune', *dois jornais* 'two newspapers'), and [z] emerges as syllable onset before vowels (e.g., *meus amigos* 'my friends'). Nevertheless, young speakers frequently use [ʒ] before vowels (e.g., *ele faz amigos* 'he makes friends'), thus, generalizing the use of this variant before [+voiced] segments (Andrade, A. & Rodrigues, C. 2004, 2005; Rodrigues, C. & Hora 2016). Sequences involving coda-/S/ and a voiceless fricative in SEP merge in fast speech, exhibiting [ʃ] in onset, e.g., *piscina* 'swimming pool', *meus chapéus* 'my hats'. A similar process occurs if the fricative is voiced. The sequence of fricative segments ends up as [ʒ], e.g., *disjunção* 'disjunction', *três jardins* 'three gardens' (Andrade, A. & Rodrigues, C. 2004, 2005).

As for voiced stops /b/, /d/, and /g/, they are often produced in SEP as non-strident fricatives [β], [ð], and [ɣ], respectively. The process of spirantization applies in un-branched and branched onsets postvocalically at the word level (e.g., *abalo* 'shake', *poder* 'power', *agonia* 'agony', *abrandar* 'to slow down', *medrar* 'to grow', *agraciar* 'to grace'). It is less clear if this process can also apply at the beginning of a word or sentence. Several authors have tried to identify the factors underlying this process (see, e.g., Andrade, A. et al. 1978; Andrade, E. d'1977; Lousada 2006; Rodrigues, C. et al. 2014; Veloso 1995; Viana 1984; among others). According to Andrade, E. d' (1977: 189), the spirantization rule in Figure 11.4 accounts for the data. The rule excludes two contexts, namely, sentence-initial and after a homorganic [-continuous] consonant, although it may also apply to word-initially depending on the speech style.

Andrade, A. et al. (1978) analyze acoustic data from participants from Lisbon and conclude that the spirantization rule that applies to /b/, /d/, /g/ occurs at different rates and may be constrained by social factors. Viana (1979:), Veloso (1995), and Lousada (2006) also suggest that all three voiced stops may devoice phonetically. Veloso (1995) further signals that the speakers use non-strident allophones more often if they do not pay attention to discourse. In the speech of Lisbon and Braga, Rodrigues, C. (2003) observed that spirantization targets all three voiced stops, but in different rates—/d/ is less often weakened than /b/ and /g/. Since only a perceptual analysis was conducted in this study, further research is required. Apart from that, the role played by social and discourse factors should also be explored.[6]

3 Phonologic and phonetic features distinguishing major dialects of EP

Continental EP has two major dialectal areas (Cintra 1971; Segura 2013), one in the North and another in the Central-South region, which includes SEP. The varieties from these dialectal areas

are mutually intelligible. The same happens with the insular EP varieties from Madeira and the Azores, although some phonetic features of insular dialects may occasionally affect speech comprehension by speakers from other varieties. Some features from Northern dialects are usually absent from other Portuguese dialects. These features are discussed in the present section, while the features that characterize other dialects are addressed in §4.

Starting with mainland dialects of EP, the most important difference between Northern EP and Central-Southern EP is the presence of /β/ in Northern dialects, instead of /v/. This phenomenon is sometimes referred to as the 'switch of b and v', as if there were evidence that Northern dialects ever had /v/. We will not refer to this phenomenon of Northern dialects this way, as the phonetic variants of /b/ and /β/ are not the same. The stop /b/ is produced as [b] and [β] (as it happens in SEP, Rodrigues, C. 2003), while /β/ is produced as [β] or [v] (Rodrigues, C. 2003). The labiodental [v] occurs in the speech of Northern speakers, mainly, in monitored speech and when speakers from Southern dialects are present. In fact, the onset of the second syllable in words such as *lavar* 'to wash', *móvel* 'mobile' is produced with [β] in Northern EP, while it is produced with [v] elsewhere. Thus, the phonological contrast between the labial stop /b/ and the non-strident fricative /β/ of Northern dialects of EP is strictly based on the feature [continuous], an inheritance of Old Portuguese (Maia 1986, cf. also Chapter 14). All Southern and a number of Central dialects distinguish /b/ from /v/, a contrast strengthened by the feature [strident]. These dialects present minimal pairs such as *bala* 'bullet' vs. *vala* 'ditch', in opposition to the Northern group of dialects, which may neutralize the difference phonetically. The presence of [β] is stigmatized in words spelled with <v> in EP, e.g., *valor* 'value', *novo* 'new', *mover* 'to move'. Northern speakers show linguistic insecurity in formal interactions, sometimes avoiding the use of [β] in words spelled with <v> and hypercorrecting in words having /b/, e.g., *a*[v]*undância* 'abundance'. This feature is, then, stereotypical of the Northern speech in EP (Rodrigues, C. 2003, 2013).

A second feature that distinguishes Northern from Southern EP is the presence of /eI/ and /oU/ produced as diphthongs. Although SEP is included in the Southern variety of EP, it diverges from most Southern dialects in the production of /eI/. In the Northern dialectal area, /eI/ allows the phonetic forms [ej] and [ɐj], while /oU/ allows [ow] or [ɐw]. The sub-areas where each phonetic form is more common are not well defined. For the moment, we hypothesize that [ej] prevails in the Porto region, while [ɐj] is produced more frequently in Trás-os-Montes and other Northern regions, based on data from young children's oral and written productions from Porto, Bragança, and Chaves (Rodrigues, C. & Gomes 2022; Rodrigues, C. & Lourenço-Gomes 2016, 2017). As for the diphthong /oU/, there are insufficient data about the spread of its variants (e.g., [ow], [o], and [ɐw]) in the Northern area. Southern EP speakers (SEP included) produce the diphthong /oU/ as [o].

The third feature that distinguishes most Northern varieties from Southern varieties is the production of apical fricative sounds. Northern dialects have either a system with /s, z, s̪, z̪/ or a system with only /s̪, z̪/. Dialectal descriptions mention that the former system is more conservative than the latter and has a direct correlation to the spelling of EP (Andrade, A. & Slifka 2006; Segura 2013). SEP simplified the four fricative consonant system into a system with two coronal fricative sounds, i.e., /s, z/, while some Northern dialects prefer a system with two apical phonemes, /s̪, z̪/.

According to Segura (2013), the fricative consonants in *c̪era* 'wax', *c̪inco* 'five', *cruz̪eiro* 'cruise', *dez̪embro* 'december', which derive historically from words with the affricates /ʦ/ and /ʤ/, are produced as [s] and [z] (coronal fricatives) in Alto Minho, Trás-os-Montes, and in parts of Beira Alta, near the Spanish border, where apical fricatives /s̪, z̪/ from Medieval Portuguese are still preserved, as in *s̪aco* 'bag', *mis̪s̪a* 'mass', *cas̪a* 'house', and *pes̪o* 'weight'. However, this system appears to be on the wane, since Martins, A. M. and Saramago (1993) find it only in two

localities in their research for the *Atlas Linguarum Europae* (recorded in 1975). In Baixo Minho, Douro, Beira Alta, and in the Northern part of Beira Baixa, the four aforementioned consonants (/s, z, s̠, z̠/) have been neutralized in favor of the apical sounds /s̠, z̠/.[7] Northern varieties of EP most often exhibit [-anterior] segments, with or without voicing, depending on the voice value of the following segment in syllable coda. Some speakers use apical sounds in word-endings, instead. In SEP, as already mentioned, /S/ corresponds to [ʃ] before voiceless consonants or at the end of a sentence, and to [ʒ] before voiced consonants. The words *pa*[ʃ]*ta*[ʃ] 'folders', *li*[ʃ]*ta* 'list', *ma*[ʃ]*cote* 'mascot', *re*[ʒ]*ma* 'ream', *pri*[ʒ]*ma* 'prism', and *Li*[ʒ]*boa* 'Lisbon' illustrate the assimilation of voicing in syllable codas. The specification of the consonant as [-anterior] also applies to /S/ at the end of the word, so it is not due to assimilation of any kind.

The fourth phonological characteristic from Northern dialects is the pronunciation of affricate [ʧ] for the grapheme <ch>, as in *chave* 'key', where SEP has [ʃ]. This characteristic has almost disappeared from EP, however. The only regions where it can still be found are rural areas of Trás-os-Montes (Aguiar Santos 2008) and of Northern Minho, besides some isolated places in the Central dialectal area (some of them near Coimbra, cf. Santos 2003).

A sub-area of the Northwestern dialect, including the area of Porto, Braga, Guimarães, and other cities near the Northern coast, exhibits a vast range of what seem to be rising diphthongs. Indeed, the process has been called 'diphthongization' in dialectal studies, even though different terms are possible, namely, palatalization and labialization, as suggested in Rodrigues, C. (2002a). The rule involves more often the spreading of the labial feature and, to a certain extent, of the coronal feature. The words *dor* [ˈdʷoɾ] 'pain' and *sede* [ˈsʲedɨ] 'thirst' illustrate this phenomenon. The rule may also apply if the syllable nucleus is nasalized (e.g., *longe* [ˈlʷõʒɨ] 'away'), in word-initial position (e.g., *hoje* [ˈʷoʒɨ] 'today'), and between words within the intonational phrase (e.g., *Vais onde?* [ˈvajˈzʷõðɨ] 'Where are you going?'). The spreading of the labial feature permitted the development of forms where the vowel loses its labiality, e.g., *todo* [ˈtʷɐðu] 'all', *porto* [ˈpʷɐɾtu] 'harbor'. These forms confirm the mobile nature of the labial feature, as it also happens with the feature [+high] in Portuguese elsewhere (as discussed below). It is worth mentioning that in the same dialectal sub-region other so-called diphthongs emerge, particularly in word-endings with stressed vowels /e/, /ɛ/, /o/ or /ɔ/ at the end of a sentence. These may exhibit [ɛɐ] as in *café* 'coffee' or [ɔɐ] as in *avó* 'grandmother', where the second part of the vowel bears the dorsal feature (cf. Brissos & Rodrigues, C. 2016, a phonetic and phonological study of the dialectal data of the Northwestern region of Portugal). In a few localities within this sub-dialectal area, /i/ may also correspond approximately to [ej], for instance, in Castelo do Neiva, e.g., *agonia* 'agony', *ali* 'there' (that is, when there is no consonant after /i/). In addition, Northwestern areas may also present other more restricted features, namely, the neutralization of /e/ and /ɛ/ as [e] in Castelo do Neiva and Arcos de Valdevez. For further details concerning the acoustic nature of vowel subsystems found in the Northwest and the geographical coverage of the features of Northwestern EP, see Brissos and Rodrigues, C. (2016).

We turn now to the features of Central-Southern dialects and insular dialects of EP.

4 Phonological phenomena of other dialects of EP

Central and Southern dialects of EP have most of the vowel features of SEP, except for the diphthongized form of /eI/, which is simplified in the Southern dialects. Furthermore, there is not much variation in the production of consonants. Although Southern dialects exhibit internal coherence

of major features, a closer look may reveal a more complex scenario, as underscored in several studies (e.g., Brissos 2014, 2016; Brissos & Saramago 2014; Segura 1987, 2013; among others). In fact, these studies have shown that, besides the two well- known restricted areas with special vowel inventories (i.e., one in the interior of Northern Alentejo and the other in the Western part of the Algarve), some of their features can be found in the entire Southern EP region and also in Central dialects (cf. Santos 1997, 2003, for the reported the presence of [y] near Coimbra). This is the case of palatalized /u/ as [y] (a true front vowel) and as [ʉ] (a central vowel), as in *fumo* [ˈfymu] 'smoke' (Brissos 2014), found sporadically in the *Portuguese Linguistic Atlas* (*Atlas Linguístico-etnográfico de Portugal e da Galiza*), such as in Praia da Salema, Foros da Casa Nova, and Alpalhão.

The sub-dialect of Alentejo has been described in detail in several studies from a dialectological viewpoint (e.g., Brissos 2012, 2014) and, in what concerns the vowel system, with acoustic analysis. Besides the presence of [y] for /u/, Brissos (2014: 66) finds that /a/ is also palatalized as [ɛ] in this dialectal area, e.g., *pisado* 'stepped on', *tosquiar* 'to shear'. The diphthong /eI/ has changed into [ø], e.g., *couro* 'leather', /e/ has changed to [œ], e.g., *caber* 'to fit', and word-endings with /o/, e.g., *cabresto* 'halter', *cesto* 'basket' (where SEP has [u]), either have [ɨ] or vowel deletion.

In the Algarve region, the sub-dialect of the Southwest presents a chain shift of the vowel system (Segura 2013). Front vowels are lowered: /i/ is lowered almost as [e] (e.g., *figo* 'fig'), /e/ as [ɛ] (e.g., *bater* 'to hit'), /ɛ/ as [æ] (e.g., *quebra* 'break'). The vowel /a/ becomes [ɒ] (e.g., *fada* 'fairy'), and labial vowels are subject to a rising rule: /ɔ/ emerges as [o̞] (e.g., *cobra* 'snake') and /u/ becomes [y] (e.g., *mudo* 'mute'). The diphthong /eI/ becomes [e], e.g., *feito* 'done'. Other properties of this variety include the following: i) some word-endings with stressed [ɒ] correspond to the diphthong [ɒw] (e.g., *pá* 'shovel'), ii) the diphthong [ɛw] becomes [æ] (e.g., *chapéu* 'hat'), iii) word-final unstressed [u] (e.g., *seco* 'dry') and [ɨ] (e.g., *pote* 'jar') are deleted. For further details, see Brissos (2014).

Insular Portuguese, spoken in the islands of Madeira and the Azores, shares most features with Central-Southern EP. Despite this similarity, there are differences concerning vowels and consonants to which we must pay attention.

The lateral coronal consonant /l/ of syllable onsets, in Madeira and some islands of the Azores (namely, Graciosa, São Jorge, and less so in some of the islands in the Central and Western Azores)[8] has developed progressive palatalization from a preceding coronal high vowel or glide, e.g., [i] (*família* 'family') or [j] (*baile* 'dance'), even if the preceding vowel is an underlying /e/ (*elétrico* 'electric'). The palatalization occurs at different rates depending on the sociolinguistic profile of speakers, and some hypercorrected forms may also be found in words such as *ilha* [ˈilɐ] 'island', *milho* [ˈmilu] 'corn' (Carvalho 2016: 48). Without acoustic analysis, this is hypothesized as a secondary articulation, i.e., [lʲ] (Andrade, E. d' 1994; Rodrigues, C. 2002b) or as full assimilation, i.e., [ʎ] (Segura 2013: 106–110). The following words illustrate the contexts wherein this process applies, namely, within the word (sometimes involving clitics) and between words: *vilão* 'villain', *família* 'family', *filosofia* 'philosophy', *papoila* 'poppy', *parti-lo* 'to break it', *vai levar* 'will take', *Helena* [iˈlʲenɐ] 'Helene'. The palatalization of the lateral consonant may give rise to forms where the [i] has been deleted in the first syllable, e.g., *filosofia* 'philosophy' [flʲuzuˈfiɐ], *vilão* [vˈlʲɐ̃w] 'villain'. If we consider that these dialects display palatalization as a progressive assimilation process, we can state that they still have the contrast between /ʎ/ in *filha* 'daughter' and /l/ in *fila* 'row', although the latter is produced as [lʲ].

Similarly, the consonant /n/ undergoes a progressive assimilation process from preceding [i]. Segura (2013: 107) provides examples of this rule from the islands of Madeira, Terceira, and some

other Azorean islands, such as *vinagre* 'vinegar', *espinafre* 'spinach', *destino* 'destiny', and *continente* 'continent'.

A peculiar process of some dialects of Madeira and the Azores affects coda-/S/, making it surface as [j] or [i] (cf. Segura 2013: 107). In the case of the Azorean Islands, this occurs in Flores, São Miguel (Mosteiros), and São Jorge (Calheta) in the speech of uneducated speakers above all. The process applies to word-final fricative codas within the phonological phrase (except before voiceless stops): e.g., *gatos grandes* 'big cats', with [i] and deletion of the morphological theme marker /o/, *as vinhas* 'the vineyards' and *fugimos dele* 'we run from him' with [j]. Coda-/S/ can also be completely assimilated if followed by a sonorant consonant in the island of São Jorge, e.g., *fazemos laços* [fɐˈzemuˈlasuʃ] 'we make loops', *faz mal* [ˈfaˈmał] 'does harm'. This nuclearization process shows a coda avoidance strategy that EP also seems to have in the words ending in /l/ (for which some authors argue that nuclearization of lateral best explains the data—see Freitas 1997 for acquisition data and González 2008 for acoustic data from adult speakers).

In addition, coda-/S/ can be produced as [ɾ] in São Jorge, Pico, and Faial (Azores), e.g., *cardos dourados* 'car[durð]ourados' 'golden thistles' (Segura 2013: 108). This process shows that it is preferable to have a sonorant than a fricative sound in the coda position in these varieties. It is worth mentioning that this process allows the preservation of the feature [+continuous] of the underlying consonant. However, this is not the most general behavior of coda /ɾ/ in the islands of Azores. This segment is more often deleted at the end of a sentence, e.g., *Ele está a cantar.* 'He is singing.'.

Another deletion process targets velar /g/ in Southern and Northwestern Madeira and in Porto Santo if it follows [i] / [u] or a glide (Segura 2013: 108). By this rule, words such as *cigarro* 'cigarette' and *fogueira* 'bonfire' become [siˈaɾu] and [fuˈɐjɾɐ]. Contrary to this process, [ɣ]-insertion takes place in the same group of Madeira dialects, which has the effect of breaking up vowel hiatus, e.g., *frio* 'cold', *arrepio* 'shiver'.

A known phenomenon of most dialects in Madeira (cf. Segura 2013) targets the stressed high vowels /i/ and /u/, which undergo a diphthongization rule deriving [ɐj] or [ij] (e.g., *espiga* 'cob', *amiga* 'friend') and [ɐw] or [ɨw], respectively (e.g., *luta* 'fight' or *fumo* 'smoke'). The phonetic falling diphthong begins with a dorsal sound (either a [+high] or a [-high, -low] vowel) and ends with a coronal/labial glide, thus strengthening the perceptive contrast of the stressed syllable. It can also apply before the nasal [-anterior] consonant /ɲ/, e.g., *varinha* [vɐˈɾɐ̃jɲɐ] 'stick' and *punho* [ˈpɨ̃wɲu] 'fist', where /i/ and /u/ were nasalized.

We could say, then, that high vowels split into two [-cons] roots associated to the syllable nucleus (a vowel and a glide sharing the same V-place values, [ij] or [uw]) and that the glide preserves the V-place features of the underlying form allowing the vowel to undergo other rules.

First, let us focus on the rules involving stressed /i/. On the one hand, in accordance with the Obligatory Contour Principle (OCP), the surface form [ɨj] emerges with the coronal /i/ changing into a dorsal high vowel, e.g., *espiga* 'esp[ɨj]ga' 'cob'. On the other hand, afterwards, the dorsal high vowel [ɨ] changes into a non-high dorsal vowel, [ɐ], e.g., *espiga* 'esp[ɐj]ga'. If the underlying stressed vowel is /u/, [ɨw] is the second step of the changing process, since the labial node is delinked from [u] deriving [ɨ]; and the third step further increases the difference between the segments of the nucleus, with the change into a non-high vowel [ɐ]. Thus, both structures undergo similar rules deriving phonetic falling diphthongs.

Another characteristic of insular dialects is the assimilation process illustrated in (3) (cf. also Segura 2013).

(3)

precisar	[pɾisiˈzʲaɾ]	‘need’
e calam	[iˈkʲalɐ̃w]	‘and shut up’
em cesto	[ĩˈsʲeʃtu]	‘in basket’
em ferro	[ĩˈfᶠɛru]	‘on iron’
em tacho	[ĩˈtʲaʃu]	‘in pan’
de trigo	[dɨˈtrig]	‘of wheat’
trocado	[truˈkʷadu]	‘exchanged’
um cesto	[ũˈsʷeʃtu]	‘the basket’
um tacho	[ũˈtʷaʃu]	‘the pan’
o berro	[uˈbʷɛru]	‘the shout’
o trigo	[uˈtrojg]	‘the wheat’

For this spreading rule of the features coronal/labial there are two proposed analyses, according to which the coronal/labial feature spreads either to the following syllable, as a glide, or to the following onset, as a secondary articulation of the consonant.

Dialectal studies describe this phenomenon as a rising diphthong rule (Segura 2013), while the phonological analysis provided by Rodrigues, C. (2002a) claims that it is a secondary articulation development—similar to the proposed analysis of the aforementioned labialization phenomena in Northwestern varieties. Both dialectal and phonological approaches rely on perceptual data alone, which is not enough to solve the controversy, since native speakers of Portuguese do not perceive differences between a labialization of a consonant and a rising diphthong that follows a consonant (Madruga & Abaurre 2015: 367). The dialectal approach seems to imply an increase of weight in the syllable, while the phonological approach does not, since labialization/palatalization is associated with the onset of the syllable. Rodrigues, C. (2002b) argues that there is not a strong reason to create rising diphthongs in EP, especially if this implies the creation of a marked structure that is unavailable in Portuguese elsewhere. As mentioned previously, only the more conservative dialects of Northwestern Portugal display so-called rising diphthongs. In fact, SEP and other major varieties of the language lack rising diphthongs (except those that derive in fast speech from a full vowel, e.g., *teatro* [ˈtjatɾu] ‘theater’, *doar* [ˈdwaɾ] ‘to donate’, and which alternate with hiatus production) and have often changed diphthongs into vowels, thus preferring unbranched nuclei. The formation of falling diphthongs is allowed only in two circumstances, namely, in the pluralization of nouns ending in /l/ in the singular form (e.g., sg. *animal* ‘animal’. pl. *animais* ‘animals’) and before palatal segments, as already mentioned.

Even though Northwestern and insular phenomena were traditionally treated in a similar way, they seem to be different. The former is a regressive spreading applying within the syllable, while the latter involves a progressive spreading of the articulator feature of the unstressed vowel to the following stressed one. Another difference is that the insular rule may apply if the nucleus on the left has a vowel or a glide independently of it being in the same word or in a preceding one, which is not the case of the other process.

In Madeira and the Azores, the process modifies words, such as *trocado* ‘exchanged’ and *ficar* ‘to stay’ into [tɾuˈkʷadu] and [fiˈkʲaɾ], respectively. This rule also applies between words (cf. Segura 2013: 109), e.g., *o/um pássaro* [uˈpʷasɐɾu]/[ũˈpʷasɐɾu] ‘the bird’, so prosodic features must be further investigated. This process also derives two forms for the same word, one after coronal vowels/glides and another after the labial vowels/glides, as it happens in *em ferro* [ĩˈfᶠɛru] ‘on iron’ and

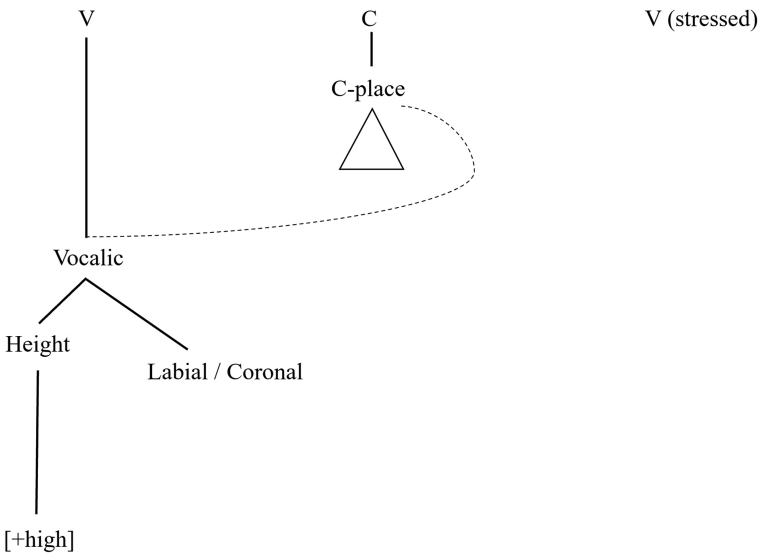

Figure 11.5 Labial / Coronal spreading.

o ferro [uˈfʷɛru] 'the iron'. The following examples show that the spreading of the labial feature may develop a falling diphthong or a complete modification of the articulation place of the target stressed vowel, if it is followed by [-anterior] coronal consonant: *o trigo* [uˈtɾujg] / [uˈtɾojg] 'the wheat' and *bolinho* [buˈluɲu] 'small cake', *o vinho* [uˈvoɲu] 'the wine', *o filho* [uˈfoʎu] 'the son'. These examples show that this assimilation process in Madeira and the Azores may go through several steps further but does not create a rising diphthong.

We may consider, then, that the process is a spreading rule involving either the labial or the coronal feature of high unstressed vowels /u/ and /i/, which develops the labialization or the palatalization of the consonant in the following stressed syllable, as shown in Figure 11.5.

When the process goes a step further, it changes the nucleus of the stressed syllable, first creating a falling diphthong and, afterwards, a stressed vowel with the place features of the vowel of the preceding syllable, e.g., *porquinho* 'por[ˈku]nho' 'little pig', *bolinho* 'bo[ˈlu]nho' 'little cake'. The [-anterior] consonants inhibit the presence of the glide (Segura 2013: 112), a fact that may be related to their complex nature in EP (they have both C-place and V-place nodes).

Both labialization and palatalization operate in Madeira and in the Azores (with the exception of the island of São Miguel), but on Terceira Island, besides being more often applied, other processes can apply cumulatively (Segura 2013: 113). To a greater or lesser degree, these processes consist of the progressive assimilation of unstressed vowel/semivowel features by the stressed vowel (e.g., *cidade* [siˈðɛðɨ] 'city' with /a/ becoming coronal [ɛ], *fumar* [fuˈmɒɾ] 'to smoke' with /a/ becoming labial [ɒ]). There is an asymmetry in the use of labialization and palatalization among Azorean islands, since only in Corvo and Flores is palatalization more productive than labialization.

The island of São Miguel exhibits phonological singularities (cf. Krämer 2017; Segura 2013), not found in any other islands nor in continental EP. One of the well-known properties of this dialect is the chain shift of the stressed vowel system (see Segura 2013: 111), whereby /i/ is lowered into [ɪ] (sometimes, even [e]), e.g., *malinha* 'mal[ɪ]nha' 'briefcase', /e/ becomes dorsal [ɜ] e.g., *tela* 't[ɜ]la' 'screen', /ɛ/ becomes [æ], e.g., *festa* 'f[æ]sta' 'party', /a/ shows several degrees of round-

ing, e.g., *focar* ‘foc[ɒ]r/foc[ɑ]r’ ‘to focus’, /ɔ/ may be produced less open as an [ɔ̝], e.g., *cobre* ‘c[ɔ̝]bre’ ‘copper’, /o/ is more closed, e.g., *lodo* ‘l[u]do’ ‘mud’, and /u/ becomes [y], e.g., *figura* ‘fig[y]ra’ ‘figure’. This dialect deletes unstressed [u] and [ɨ] in word-final position, e.g., *pente* ‘comb’ or *gato* ‘cat’. The diphthong /eI/ monophthongizes to [e] (e.g., *leite* ‘l[e]te’ ‘milk’), while /oU/ and /oI/ do so to [ø] (e.g., *falou* ‘fal[ø]’ ‘spoke’, *loiça* ‘l[ø]ça’ ‘crockery’).

5 Concluding remarks

This chapter covered several phonological phenomena of EP, not only in its standard variety, but also in other less studied dialects. We presented the general phonological properties of SEP, namely its phonological system and the rules undergone by vowels and consonants. We observed that SEP exhibits variation with a few segments, for instance, unstressed word-initial [–high] coronal and labial vowels, and stressed /e/ before palatal segments. It has a regressive nasalization process derived from /N/, but not a nasalization caused by nasal onsets. We have also discussed the variation concerning coda-/S/ when it is re-syllabified as onset before vowels, the variation associated with oral stops, and the sonorant /ɾ/ in syllable onset and syllable coda. Further research on each of these phenomena is needed, from a phonetic and a phonological perspective. We described most of the processes found in EP in auto-segmental terms for convenience, most of the time referring only to the word level. However, some processes require further research since higher prosodic factors may have an impact.

The most relevant features that differentiate Northern EP dialects from SEP were presented in §2. We addressed not only the best-known features of different dialects of EP, but also a set of phenomena that have received recent attention in the literature and that are found in a larger area of Northwestern Portugal than has been traditionally admitted by dialectal studies. Northern dialects show several properties that may be considered as conservative features (e.g., /β/ instead of /v/, the presence of the affricate /ʧ/ or the presence of apical fricatives), although they are also characterized by innovation, which merits future investigation, such as the behavior of the diphthong /eI/.

Next, we presented a selection of features of Central-Southern dialects, including some that characterize their sub-dialects (one in the extreme South of the Algarve and the other in the Northeastern part of the Alentejo region). Dialects in this wide region share most features with SEP, such as the simplification of the diphthong /oU/, as well as the same consonantal system. However, they go further than SEP in the simplification of the diphthong /eI/. Some features previously assumed to be present only in a few places of this geographical area have recently been shown to be present in the entire Southern area. The two dialectal sub-areas of the Central and Southern dialects are recognized by the complexity of their vowel inventories. Although both of them share some properties, we have called attention to their main differences (for instance, there is evidence of a chain shift in Barlavento of Algarve that is absent from the sub-dialect of Alentejo, near the Spanish border).

In addition, we described a few phonological phenomena of insular EP that still need to be further investigated. These phonological and phonetic phenomena show the vitality of EP, especially in rural and isolated communities, as it often happens in the islands of Madeira and the Azores. Although one may find similarities among all the insular dialects, there are usually certain properties that are unique to each one of them. Even if they share most features with Central-Southern EP (including the sub-dialects of Barlavento and Alentejo), they took on separate paths in many aspects, as with the development of secondary articulations of consonants and the development of the vowel chain shift in the island of São Miguel. While it is impossible to cover the entire range of variable segmental processes in EP, the diversity of phenomena described in this chapter is meant

to encourage new research and collaboration among scholars so as to deepen our understanding of some of the more complex issues in EP phonology. Most of the aforementioned processes still lack acoustic analysis, sociolinguistic scrutiny, and studies investigating the prosodic environments where they may and may not apply, which we consider of major importance for future research.

Notes

1 Saying this does not mean that some of the features discussed here, such as the Coronal / Labial spreading of insular Portuguese discussed below, are not constrained by higher prosodic domains. We decided not to discuss other phonological (or phonetic) features of EP, such as those targeting word-final unstressed vowels and other sandhi phenomena, namely the word-final [j] insertion and other unstressed vowel deletion, which all apply only in certain higher prosodic domains in fast speech. We encourage the reader to resort to further readings to improve their knowledge of these features. For instance, Rodrigues, C. (2003) reports on unstressed /e/ and /o/ deletion within the word, e.g., *prefe̲rir* [pɾɨfˈɾiɾ], 'prefer' *de̲finir* [dfˈniɾ] 'define' and *do̲cumento* [dkuˈmẽtu] 'document', *co̲zinha* [kˈzĩɲɐ] 'kitchen', which occurs quite often in fast speech. Segura (2013) and Oliveira et al. (2017) report on glide-insertion to break a hiatus in sandhi, e.g., *a* [j] *água* 'the water', which occurs in the Northern dialects of EP. Rodrigues, C. (2015) and Cruz (2013) report the insertion of [i] / [j] or [ɨ] after sonorant (/n/ or /l/) word-final codas in words ending in a stressed syllable, e.g., *andar* 'walk', *sal* 'salt', or with a final open syllable, e.g., *café* 'coffee', a common rule of Central and Southern dialects of EP.

2 There are also other historical allophonic rules. Nevertheless, since they have already been lexicalized, we will not include them here (for instance, the phonological alternation that targeted the voiced velar stop at the end of a morpheme followed by /i/, e.g., *demagogo* [dɨmɐˈɣoɣu] 'demagogue' / *demagogia* [dɨmɐɣuˈʒiɐ] 'demagogy', as all speakers of Portuguese know when [ʒ] is expected). These will not be discussed in this chapter due to the presence of different consonants in the underlying form of the words (for further discussion, see Andrade, A. 2020; Rodrigues, C. 2020).

3 Within the phonological tradition of SEP, the phonetic symbol [ɐ] (a central mid vowel) may correspond to an underlying /a/ in unstressed syllables (e.g., *a̲zedo* 'sour', *a̲té* 'until') or to an /e/ in stressed syllables before palatal sounds (e.g., *abe̲lha* 'bee', *cere̲ja* 'cherry'). The phonetic symbol [ɨ] corresponds to a central high vowel in unstressed syllables with underlying /e/ or /ɛ/ (e.g., *se̲lado* 'sealed', *sede̲* 'thirst', *fe̲bril* 'feverish', *ne̲blina* 'fog').

4 We do not intend to describe the stress patterns of EP here, but the reader may be interested to read, for instance, Pereira, I. (2020) and Lu et al. (2023), from which we can conclude that the most common stress pattern in Portuguese is a trochaic one, i.e., with the stress falling on the penultimate syllable, e.g., *amare̲lo* 'yellow'. The other two patterns are less frequent, namely, with the stress falling on the last syllable (e.g., *jacare̲é* 'alligator') and on the antepenultimate syllable (e.g., *mú̲sica* 'music').

5 Rodrigues C. (2003) concludes that rhotic is subject to variation in onset position in Lisbon, although [r] only emerges in the speech of some old speakers.

6 Rodrigues, C. et al. (2014) analyzed the speech of eight Braga speakers, concluding that the spirantization rule applies often.

7 See Martins, A. M. and Saramago (1993) and Andrade, A. and Slifka (2006) for more details on the acoustic properties of apical sounds in EP.

8 In the island of São Miguel (Azores), /l/ is normally produced as an apical sound.

Further reading

Cruz, Marisa. 2013. *Prosodic Variation in European Portuguese: Phrasing, Intonation, and Rhythm in Central-Southern Varieties*. Ph.D dissertation, University of Lisbon.

Oliveira, Pedro, Marisa Cruz, Nuno Paulino, and Marina Vigário. 2017. Glide insertion to break a hiatus across words in European Portuguese. In Pilar Barbosa, Maria da Conceição Paiva, and Celeste Rodrigues (eds.), *Studies on Variation in Portuguese*. Amsterdam/Philadelphia: John Benjamins, 49–79.

Saramago, João. 1992. *Le parler de l'île de Corvo - Açores*. Grenoble: Université Stendhal-Grenoble III / Center of Linguistics of the University of Lisbon.

Segura, Luísa. 1991. *O Falar de Odeleite*. Lisbon: Instituto Nacional de Investigação Científica.

References

Andrade, Amália. 2020. Vocalismo. In Eduardo Raposo, Fernanda Bacelar Nascimento, M. Antónia Mota, Luísa Segura, Amália Mendes, and Amália Andrade (eds.), *Gramática do Português*, vol. 3. Lisbon: Fundação Calouste Gulbenkian, 3241–3332.

Andrade, Amália and Celeste Rodrigues. 2004. Um exemplo de sandhi consonântico variável em português. In *Actas do XIX Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 257–268.

Andrade, Amália and Celeste Rodrigues. 2005. Fusão de sibilantes: Um processo de mudança/standardização? *Actas do XX Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 363–371.

Andrade, Amália, Ernesto d'Andrade, and Maria do Céu Viana. 1978. *A fricatização das oclusivas sonoras em português*. Relatórios do Grupo de Fonética e Fonologia – 3. University of Lisbon.

Andrade, Amália and Janet Slifka. 2006. Estudo fonético de sibilantes produzidas por 2 falantes de um dialecto português setentrional. In Fátima Oliveira and Joaquim Barbosa (eds.), *Textos seleccionados do XXI Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação de Linguística Portuguesa, 109–123.

Andrade, Ernesto d'. 1994. Algumas particularidades do português falado no Funchal. In *Actas do IX Encontro Nacional da Associação Portuguesa de Linguística*. Coimbra, Lisbon: Associação Portuguesa de Linguística, 17–29.

Andrade, Ernesto d'. 1977. *Aspects de la phonologie (générative) du portugais*. Lisbon: Instituto Nacional de Investigação Científica.

Andrade, Ernesto d' and Maria do Céu Viana. 1993. Que horas são às (1)3 e 15? In *Actas do VIII Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 59–66.

Andrade, Ernesto d' and Maria do Céu Viana. 1994. Sinérese, diérese e estrutura silábica. In *Actas do IX Encontro Nacional da Associação Portuguesa de Linguística*. Lisboa: Associação Portuguesa de Linguística, 31–42.

Aguiar, Santos and Isabel Joana. 2008. *Unidades e processos fonológicos no falar da região da Terra Quente: Contributos para a linguística forense*. Master's thesis, University of Minho.

Atlas Linguarum Europae—ALE. 1983–1990. I, 1–4 Assen-Maastricht: Van Gorcum; 1998. I, 5 Roma: Istituto Poligrafico e Zecca dello Stato; (2002) I, 6 Roma: Istituto poligrafico e Zecca dello Stato; 2007. I, 7 Roma: Istituto Poligrafico e Zecca dello Stato.

Atlas Linguístico-Etnográfico de Portugal e da Galiza—ALEPG. Lisbon: University of Lisbon.

Archangeli, Diana. 1988. Aspects of Underspecification theory. *Phonology* 5.183–208.

Barros, Rita Queirós de. 1994. *Contributo para uma análise sociolinguística do português de Lisboa: Variantes de /e/ e /ɛ/ em contexto pré-palatal*. Master's thesis, University of Lisbon.

Bonet, Eulália and Joan Mascaró. 1997. On the representation of contrasting rhotics. In Fernando Martínez-Gil and Alfonso Morales-Front (eds.), *Issues in the Phonology and Morphology of the Major Iberian Languages*. Washington, DC: Georgetown University Press, 103–126.

Brissos, Fernando. 2012. *Linguagem do Sueste da Beira no tempo e no espaço*. Lisbon: University of Lisbon.

Brissos, Fernando. 2014. New insights into Portuguese central-southern dialects: Understanding their present and past forms through acoustic data from stressed vowels. *Journal of Portuguese Linguistics* 13.63–115.

Brissos, Fernando. 2016. Portugal: A cidade e o interior. I – Centro-sul. *Limite Revista de Estudios Portugueses y de la Lusofonía* 10.85–106.

Brissos, Fernando and Celeste Rodrigues. 2016. Vocalismo Acentuado do Noroeste Português – Descrição acústica, variação dialectal e representação fonológica. *Revue Romane* 51.1–35.

Brissos, Fernando and João Saramago. 2014. O problema da diversidade dialectal do Centro-Sul português: Informação perceptiva versus informação acústica. *Estudos de Lingüística Galega* 6.53–80.

Carvalho, Nélio Nunes. 2016. *Conhecimento e uso do dialeto/falar madeirense dos jovens em escolarização da Região Autónoma da Madeira - Análise de enunciados dialetais: Léxico*. Master's thesis, Open University.

Cintra, Luís F. Lindley. 1971. Nova proposta de classificação dos dialectos galego-portugueses. *Boletim de Filologia* 22.81–116.

Clements, George Nick and Elizabeth Hume. 1995. The Internal Structure of Speech Sounds. In John Goldsmith (ed.), *The Handbook of Phonological Theory*. Malden, MA: Blackwell, 245–306.

Cruz, Marisa. 2013. *Prosodic Variation in European Portuguese: Phrasing, Intonation and Rhythm in Central-Southern Varieties*. Ph.D. dissertation, University of Lisbon.

Freitas, Maria João. 1997. *Aquisição da estrutura silábica do português europeu*. Ph.D. dissertation, University of Lisbon.

Freitas, Maria João. 2001. Os pinguins são diferentes dos coelhos? Questões sobre oclusivas velares, semi-vogais e arredondamentos na aquisição do Português Europeu. In *Actas do XVI Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 213–225.

Goldsmith, John A. (ed.). 1995. *The Handbook of Phonological Theory*. Cambridge, MA: Blackwell.

González, Marcos Garcia. 2008. *Português europeu e galego: Estudo fonético e fonológico das consoantes em rima medial*. Master's thesis, University of Lisbon.

Jesus, Luís and Christine H. Shadle, 2005. Acoustic analysis of European Portuguese uvular [χ, ʁ] and voiceless tapped alveolar [ɾ̥] fricatives. *Journal of the International Phonetic Association* 35.27–44.

Krämer, Philipp, 2017. Front rounded vowels in Azorean Portuguese: A reappraisal. *Journal of Portuguese Linguistics* 16.5.

Lao, Shanyi. 2020. *Variação da nasalização regressiva heterossilábica em Portugal Continental*. Master's thesis, University of Lisbon.

Lousada, Marisa. 2006. *Estudo da Produção de oclusivas do Português Europeu*. Master's thesis, University of Aveiro.

Lu, Shuang, Cátia Severino, Marina Vigário, and Sónia Frota. 2023. Language-specific stress discrimination by European-learning infants – An ERP study. In Radek Skarnitzl and Jan Volín (eds.), *Proceedings of the 20th International Congress of Phonetic Sciences*, Prague: Guarant International, 426–430.

Madruga, Magnum and Bernardete Abaurre. 2015. Onset phonotactic restrictions and rising diphthongs in Portuguese. *Revista da ABRALIN* 14.339–379.

Maia, Clarinda de Azevedo. 1986. *História do galego-português*. Coimbra: Instituto Nacional de Investigação Científica.

Martins, Ana Maria and João Saramago. 1993. As sibilantes em português: Um estudo de geografia linguística e de fonética experimental. In Ramón Lorenzo (ed.), *Actas do XIX Congreso Internacional de Lingüística e Filoloxía Románicas*. A Coruña: Fundación Pedro Barrié de la Maza / Conde de Fenosa, 4.121–142.

Mascarenhas, Isabel. 1996. *Estudo da variação dialectal entre Lisboa e Porto das vogais átonas [-rec] e [+arred] em contexto inicial*. Master's thesis, University of Lisbon.

Mateus, Maria Helena Mira. 1975[1982]. *Aspectos da fonologia portuguesa*. 2nd edn. Lisbon: Instituto Nacional de Investigação Científica.

Mateus, Maria Helena Mira and Celeste Rodrigues. 2004. A vibrante em coda em português europeu. In *Actas do XIX Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 289–299.

Mateus, Maria Helena Mira and Ernesto d'Andrade. 2000. *The Phonology of Portuguese*. Oxford: Oxford University Press.

Oliveira, Pedro, Marisa Cruz, Nuno Paulino, and Marina Vigário. 2017. Glide insertion to break a hiatus across words in European Portuguese. In Pilar Barbosa, Maria da Conceição Paiva, and Celeste Rodrigues (eds.), *Studies on Variation in Portuguese*. Amsterdam/Philadelphia: John Benjamins, 49–79.

Pereira, Isabel. 2020. Acento de palavra. In Eduardo Raposo, Fernanda Bacelar Nascimento, Maria Antónia Mota, Luísa Segura, Amália Mendes, and Amália Andrade (eds.), *Gramática do Português*, vol. 3. Lisbon: Calouste Gulbenkian Foundation, 3399–3428.

Pereira, Rodrigo. 2020. *O rótico /ʁ/ em português europeu: Análise fonológica de dados dialetais*. Master's thesis, University of Lisbon.

Raimy, Eric, and Charles E. Cairns (eds.). 2015. *The Segment in Phonetics and Phonology*. Malden, MA: Wiley Blackwell.

Rennicke, Iliris, and Pedro Tiago Martins. 2013. As realizações fonéticas de /R/ em português europeu: Análise de um corpus dialetal e implicações no sistema fonológico. In Fátima Silva, Isabel Falé, and Isabel Pereira (eds.), *Textos selecionados do XXVIII Encontro Nacional da Associação Portuguesa de Linguística*. Coimbra: Associação Portuguesa de Linguística, 509–523.

Rodrigues, Celeste. 2002a. Questões de espraiamento em PE. In *Actas do XVII Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 419–432.

Rodrigues, Celeste. 2002b. Variação linguística em Porto. In *Actas do Encontro Comemorativo dos 25 anos do Centro de Linguística da Universidade do Porto*, vol. 1. Porto: University of Porto, 119–130.

Rodrigues, Celeste. 2003. *Lisboa e Braga: Fonologia e variação*. Lisbon: Calouste Gulbenkian Foundation.

Rodrigues, Celeste. 2013. Braga: O frágil equilíbrio entre preservação dialetal e standardização. In Ana Gabriela Macedo, Carlos Mendes Sousa, and Vítor Moura (eds.), *XIV Colóquio de Outono – Humanidades*: *Novos Paradigmas do Conhecimento e da Investigação*. Braga: Humus, 255–272.

Rodrigues, Celeste. 2015. Evidências de regularização acentual no Litoral Alentejano. *Revista da ABRALIN*, 14.463–479.

Rodrigues, Celeste. 2016. Variação sociolinguística. In Ana Maria Martins and Ernestina Carrilho (eds.), *Manual de Linguística Portuguesa*. Berlin: Mouton de Gruyter, 98–115.

Rodrigues, Celeste. 2020. Consonantismo. In Eduardo Raposo, Fernanda Bacelar Nascimento, Maria Antónia Mota, Luísa Segura, Amália Mendes, and Amália Andrade (eds.), *Gramática do Português*, vol. 3. Lisbon: Calouste Gulbenkian Foundation, 3333–3368.

Rodrigues, Celeste and Maria do Carmo Lourenço-Gomes. 2016. Estudo longitudinal da proficiência ortográfica no 2.º e 4.º anos de escolaridade – estruturas /e/, /eI/ e /oU/. *Revista Diacrítica* 30.115–136.

Rodrigues, Celeste and Maria do Carmo Lourenço-Gomes. 2017. Aprender com o erro, ensinar sem erro. Paper presented at the A Linguística na Formação do Professor Conference. University of Porto, Porto.

Rodrigues, Celeste, and Jéssica Gomes. 2022. O *quaijo* que *xerava* bem: Um estudo de produções escritas e orais do ditongo <ei> de crianças de dois dialetos portugueses. *Linguística – Revista de Estudos Linguísticos da Universidade do Porto*, Número Especial in Honorem Fátima Oliveira, 313–339.

Rodrigues, Celeste and Dermeval da Hora. 2016. Main current processes of phonological variation. In W. Leo Wetzels, Sergio Menuzzi, and João Costa (eds.), *The handbook of Portuguese linguistics*. Malden, MA: Wiley Blackwell, 504–525.

Rodrigues, Celeste, Anabela Rato, Henrique Barroso, Patrícia Varanda, Micaela Aguiar, and Vera Barbosa. 2014. A realização das oclusivas vozeadas em falantes do PSFB. Paper presented at the I International Symposium on Variation in Portuguese. University of Minho, Braga.

Rodrigues, Susana. 2015. *Caracterização acústica das consoantes líquidas do Português Europeu*. Ph.D. dissertation, University of Lisbon.

Rodrigues, Susana, Fernando Martins, and Luís Jesus. 2016. Rhotics in European Portuguese: The variability in phonetic realisations. Poster presented at the R-atics 5. Fryske Akademy, Leeuwarden, Netherlands.

Santos, Isabel Almeida. 1997. Fenómenos de palatalização vocálica na Romênia: Valor dialectal em território português (continental). In *Actas do XII Encontro Nacional da Associação Portuguesa de Linguística*. Lisbon: Associação Portuguesa de Linguística, 273–287.

Santos, Isabel Almeida. 2003. *Variação linguística em espaço rural – A vogal [ü] numa comunidade do Baixo Mondego*. Lisbon: Imprensa Nacional Casa da Moeda.

Segura, Luísa. 1987. *A fronteira dialectal do Barlavento do Algarve*. Ph.D. dissertation, University of Lisbon.

Segura, Luísa. 2013. Variedades dialectais do português europeu. In Eduardo Raposo, Fernanda Bacelar Nascimento, Maria Antónia Mota, Luísa Segura, and Amália Mendes (eds.), *Gramática do Português*, vol. 1. Lisbon: Calouste Gulbenkian Foundation, 85–144.

Teyssier, Paul. 1980. *Histoire de la langue portugaise*. Paris: Press Universitaires de France.

Veloso, João. 1995. *Aspectos da percepção das "oclusivas fricatizadas" do português*. Prova de Aptidão Pedagógica Capacidade Científica, University of Porto.

Veloso, João. 2013. Redução do vocalismo átono do português europeu contemporâneo: Assimetria dos elementos de tonalidade e interação entre diversos tipos de redução vocálica. In *Textos Selecionados do XXVIII Encontro Nacional da Associação Portuguesa de Linguística*. Coimbra: Associação Portuguese de Linguística, 655–672.

Veloso, João. 2015. The English r coming! The never-ending story of Portuguese rhotics. In Alberto Simões, Anabela Barreiro, Diana Santos, Rui Sousa-Silva, and Stella E. O. Tagnin (eds.), *Linguística, informática e tradução: Mundos que se cruzam*, vol. 7. Oslo: Studies in Language, 323–336.

Veloso, João. 2016. O sistema vocálico e a redução e neutralização das vogais átonas em português europeu contemporâneo. In Ana Maria Martins and Ernestina Carrilho (eds.). *Manual de Linguística Portuguesa*. Berlin: De Gruyter, 636–662.

Veloso, João. 2019. Complex segments in Portuguese: The unbearable heaviness of being palatal. In Irantzu Epelde Zendoia and Oroitza Jauregi Nazabal (eds.), *Bihotz ahots M. L. Oñederra irakaslearen omenez*. Bilbao: Universidad del País Vasco/Euskal Heriko Unibertsitatea, 513–526.

Viana, Maria do Céu. 1979. *Etudes sur la perception du voisement des occlusives portugais*. Paris: Institut d'Études Linguistiques et Phonétiques.

Viana, Maria do Céu. 1984. *Étude de deux aspects du consonantisme du portugais: Fricatisation et dévoisement*. Ph.D. dissertation, University of Strasbourg.

Vigário, Marina. 2003. *The prosodic word in European Portuguese*. Berlin/New York: Mouton de Gruyter.